Handwritten Digit Recognition using Neural Networks

Project Synopsis: Naviotech Solution AI Internship (September 2026 batch)

Sangini Agrawal

October 2026

Introduction

Handwritten digits appear on cheques, postal envelopes, examination sheets and paper forms. In this project, neural networks are trained in TensorFlow/Keras to recognise handwritten digits (0–9) from 28 × 28 grayscale images of the MNIST dataset, and a prediction system is built that can read a user's own handwriting.

Problem Statement

Given a 28 × 28 grayscale image of a single handwritten digit, automatically identify the digit with more than 95% accuracy and let users test the system with their own drawn or photographed digits.

Objectives

1. Understand the fundamentals of neural networks.
2. Learn image classification techniques.
3. Train a model using the MNIST dataset.
4. Evaluate model accuracy and performance.
5. Deploy a simple digit recognition system.

Dataset

Property	Value
Dataset	MNIST handwritten digits (LeCun, Cortes & Burges)
Images	70,000 grayscale, 28 × 28 pixels
Split used	54,000 training · 6,000 validation · 10,000 test
Classes	10 digits (0–9), roughly balanced

Methodology

Phase	Work done
1. Data collection	Load MNIST with Keras; explore samples, class balance, average digits
2. Preprocessing	Normalise pixels to 0–1; stratified validation split; reshape for CNN
3. Model development	Dense network 784 → 128 → 64 → 10 (ReLU,

Phase	Work done
	dropout, softmax); CNN with two Conv–Pool blocks
4. Training	Adam (lr 0.001), sparse categorical cross-entropy, batch 128, early stopping
5. Evaluation	Accuracy, macro precision/recall/F1, confusion matrices, error analysis, classical baselines
6. Prediction	MNIST-style preprocessing; draw in Colab, upload a photo, command-line tool, browser demo

Tools and Technologies

Python · TensorFlow / Keras · NumPy · Pandas · Matplotlib · scikit-learn · Pillow · Google Colab / Jupyter · Git and GitHub · HTML/JavaScript

Results

Model	Parameters	Test accuracy	Macro F1
Dense neural network	109,386	97.93%	97.91%
Convolutional neural network	34,826	99.19%	99.18%
K-nearest neighbours (k=3)	–	96.99%	–
Random forest (100 trees)	–	96.96%	–
Logistic regression	–	92.45%	–

The CNN made 81 mistakes on the 10,000 test images, against 207 for the dense network. With MNIST-style preprocessing it read all eleven sample user images correctly, as well as digits drawn and uploaded in Colab. On 30 harder photo-style images it got 16 right, compared with only 4 with a plain resize.

Applications

Bank cheque processing · postal code recognition · OCR systems · educational tools · automated form processing.

Conclusion

Both neural networks exceed the 95% target. The CNN is the most accurate (99.19%) while using about three times fewer parameters than the dense network, and most of its remaining errors are on digits that look alike and are made with low confidence. Small, faint digits in noisy photos are still a weakness. Future work includes data augmentation, deeper CNNs, multi-digit recognition and a mobile app.